

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408984
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Bono; Peter L. et al.

Surgical image system and method

Abstract

A surgical system and method are provided that is configured to effect substituting a high resolution image created pre-surgery for a lower resolution image created in real time during surgery and simulating the real time position of the surgical site.

Inventors: Bono; Peter L. (Bingham Farms, MI), Lark; James D. (West Bloomfield, MI), Scales; John S. (Ann Arbor, MI), Lord; Thomas J. (South Milwaukee, WI)

Applicant: GLOBUS MEDICAL, INC. (Audubon, PA)

Family ID: 80002365

Assignee: Globus Medical, Inc. (Audubon, PA)

Appl. No.: 17/461151

Filed: August 30, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220031395 A1	Feb. 03, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 15816861 20171117 US 11135026 child-doc US 17461151
us-provisional-application US 63142716 20210128
us-provisional-application US 62423677 20161117

Publication Classification

Int. Cl.: A61B34/20 (20160101); A61B8/12 (20060101); A61B34/10 (20160101); A61B34/30 (20160101); A61B90/00 (20160101); G06T7/10 (20170101)

U.S. Cl.:

CPC **A61B34/10** (20160201); **A61B8/12** (20130101); **A61B34/20** (20160201); **A61B34/30** (20160201); **A61B90/37** (20160201); **G06T7/10** (20170101); A61B2034/105 (20160201); A61B2034/107 (20160201); A61B2034/2051 (20160201); A61B2090/3762 (20160201); A61B2090/378 (20160201); G06T2207/10081 (20130101); G06T2207/10136 (20130101); G06T2207/30008 (20130101)

Field of Classification Search

CPC: A61B (90/37); A61B (2090/364); A61B (2090/378); A61B (2090/3893); A61B (2034/101); A61B (2034/107); A61B (2034/303); A61B (2034/363); A61B (2034/2051); A61B (2034/2063); A61B (5/066); A61B (34/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1154159	12/1914	Ashworth	N/A	N/A
2557429	12/1950	Hawley	N/A	N/A
2831295	12/1957	Weiss	N/A	N/A
3091060	12/1962	Giegerich et al.	N/A	N/A
3554197	12/1970	Dobbie	N/A	N/A
3577579	12/1970	Duve et al.	N/A	N/A
4007528	12/1976	Shea et al.	N/A	N/A
4008720	12/1976	Brinckmann et al.	N/A	N/A
4081704	12/1977	Vassos et al.	N/A	N/A
RE29736	12/1977	Shea et al.	N/A	N/A
D248967	12/1977	Shea et al.	N/A	N/A
4111208	12/1977	Leuenberger	N/A	N/A
4596243	12/1985	Bray	N/A	N/A
4620539	12/1985	Andrews et al.	N/A	N/A
4828052	12/1988	Duran et al.	N/A	N/A
4932935	12/1989	Swartz	N/A	N/A
5092875	12/1991	McLees	N/A	N/A
5522829	12/1995	Michalos	N/A	N/A
5733119	12/1997	Carr	N/A	N/A
5843110	12/1997	Dross et al.	N/A	N/A
6021538	12/1999	Kressner et al.	N/A	N/A
6110174	12/1999	Nichter	N/A	N/A
6522906	12/2002	Salisbury, Jr.	600/407	A61B 1/0005
6546279	12/2002	Bova et al.	N/A	N/A
6606539	12/2002	Raab	N/A	N/A
6635067	12/2002	Norman	N/A	N/A
6689087	12/2003	Pal et al.	N/A	N/A
6716215	12/2003	David et al.	N/A	N/A
6721986	12/2003	Zhuan	N/A	N/A
6895305	12/2004	Lathan et al.	N/A	N/A

6966912	12/2004	Michelson	N/A	N/A
7107090	12/2005	Salisbury, Jr.	901/14	G16H 40/63
7160304	12/2006	Michelson	N/A	N/A
7194120	12/2006	Wicker et al.	N/A	N/A
7346417	12/2007	Luth et al.	N/A	N/A
7660623	12/2009	Hunter et al.	N/A	N/A
7922720	12/2010	May et al.	N/A	N/A
8029523	12/2010	Wallis et al.	N/A	N/A
8038630	12/2010	Pal et al.	N/A	N/A
8170717	12/2011	Sutherland et al.	N/A	N/A
8219178	12/2011	Smith et al.	N/A	N/A
8465491	12/2012	Yedlicka et al.	N/A	N/A
8491603	12/2012	Yeung et al.	N/A	N/A
8657821	12/2013	Palermo	N/A	N/A
8728085	12/2013	Marsh et al.	N/A	N/A
8828001	12/2013	Stearns et al.	N/A	N/A
8943634	12/2014	Sokol et al.	N/A	N/A
9101397	12/2014	Guthart	N/A	A61B 34/30
9232984	12/2015	Guthart	N/A	A61B 34/37
10271909	12/2018	Guthart	N/A	A61B 34/37
10748327	12/2019	Paulson	N/A	G06F 30/00
11259870	12/2021	DiMaio	N/A	A61B 34/71
11877897	12/2023	Shelton, IV	N/A	A61B 90/90
2003/0060927	12/2002	Gerbi et al.	N/A	N/A
2004/0050603	12/2003	Jaeger	N/A	N/A
2004/0147934	12/2003	Kiester	N/A	N/A
2005/0027397	12/2004	Niemeyer	N/A	N/A
2005/0043718	12/2004	Madhani et al.	N/A	N/A
2005/0171557	12/2004	Shoham	N/A	N/A
2005/0283175	12/2004	Tanner et al.	N/A	N/A
2006/0229624	12/2005	May et al.	N/A	N/A
2006/0235305	12/2005	Cotter et al.	N/A	N/A
2006/0235306	12/2005	Cotter et al.	N/A	N/A
2007/0005045	12/2006	Mintz et al.	N/A	N/A
2007/0021738	12/2006	Hasser et al.	N/A	N/A
2007/0276423	12/2006	Green	N/A	N/A
2007/0282344	12/2006	Yedlicka et al.	N/A	N/A
2007/0282345	12/2006	Yedlicka et al.	N/A	N/A
2008/0027449	12/2007	Gundlapalli et al.	N/A	N/A
2008/0061784	12/2007	Pal et al.	N/A	N/A
2008/0108010	12/2007	Wang	N/A	N/A
2008/0108991	12/2007	von Jako	N/A	N/A
2008/0213899	12/2007	Olgac	N/A	N/A
2010/0145343	12/2009	Johnson et al.	N/A	N/A
2010/0165793	12/2009	Haug	N/A	N/A
2010/0198230	12/2009	Shoham	N/A	N/A
2010/0249506	12/2009	Prisco	N/A	N/A
2010/0249786	12/2009	Schmieding et al.	N/A	N/A
2011/0015635	12/2010	Aryan	N/A	N/A
2011/0015649	12/2010	Anvari et al.	N/A	N/A

2011/0118708	12/2010	Burbank et al.	N/A	N/A
2011/0118709	12/2010	Burbank	N/A	N/A
2011/0118778	12/2010	Burbank	N/A	N/A
2011/0196404	12/2010	Dietz et al.	N/A	N/A
2011/0230886	12/2010	Gustilo et al.	N/A	N/A
2011/0295270	12/2010	Giordano et al.	N/A	N/A
2011/0306873	12/2010	Shenai et al.	N/A	N/A
2011/0313428	12/2010	Mohr et al.	N/A	N/A
2011/0319941	12/2010	Bar et al.	N/A	N/A
2012/0059392	12/2011	Diolaiti	N/A	N/A
2012/0186372	12/2011	Smith et al.	N/A	N/A
2012/0211546	12/2011	Shelton, IV	N/A	N/A
2012/0220831	12/2011	Cooper et al.	N/A	N/A
2012/0266442	12/2011	Rogers et al.	N/A	N/A
2013/0096540	12/2012	Cooper et al.	N/A	N/A
2013/0123799	12/2012	Smith et al.	N/A	N/A
2013/0178856	12/2012	Ye et al.	N/A	N/A
2013/0206441	12/2012	Roser et al.	N/A	N/A
2013/0244820	12/2012	Solomon et al.	N/A	N/A
2013/0245629	12/2012	Xie	N/A	N/A
2013/0296886	12/2012	Green	N/A	N/A
2013/0304069	12/2012	Bono et al.	N/A	N/A
2013/0345718	12/2012	Crawford et al.	N/A	N/A
2014/0051922	12/2013	Guthart et al.	N/A	N/A
2014/0100574	12/2013	Bono et al.	N/A	N/A
2014/0194894	12/2013	Dachs, II et al.	N/A	N/A
2014/0222003	12/2013	Herndon et al.	N/A	N/A
2014/0350391	12/2013	Prisco et al.	N/A	N/A
2014/0350571	12/2013	Maillet et al.	N/A	N/A
2015/0119916	12/2014	Dietz et al.	N/A	N/A
2016/0151120	12/2015	Kostrzewski et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
42807	12/2004	AR	N/A
370608	12/1982	AT	N/A
2011215901	12/2002	AU	N/A
2003200831	12/2003	AU	N/A
861446	12/1977	BE	N/A
1112970	12/1980	CA	N/A
2513071	12/2003	CA	N/A
2788918	12/2010	CA	N/A
610753	12/1978	CH	N/A
252004	12/2004	CL	N/A
102781349	12/2011	CN	N/A
2730227	12/1979	DE	N/A
570977	12/1977	DK	N/A
0148304	12/1984	EP	N/A
0261260	12/1987	EP	N/A
1571581	12/2004	EP	N/A

1041918	12/2005	EP	N/A
1581374	12/2005	EP	N/A
1690649	12/2005	EP	N/A
2533703	12/2011	EP	N/A
465719	12/1979	ES	N/A
773650	12/1977	FI	N/A
2374886	12/1977	FR	N/A
1550577	12/1978	GB	N/A
1081824	12/1984	IT	N/A
2006512954	12/2005	JP	N/A
4481173	12/2009	JP	N/A
2013519434	12/2012	JP	N/A
S5380789	12/2013	JP	N/A
IPS5613462	12/2013	JP	N/A
5826771	12/2014	JP	N/A
7713563	12/1977	NL	N/A
774411	12/1977	NO	N/A
WO9107116	12/1990	WO	N/A
WO0215799	12/2001	WO	N/A
WO2004062863	12/2003	WO	N/A
WO2007008703	12/2006	WO	N/A
WO2009151926	12/2008	WO	N/A
WO2011100313	12/2010	WO	N/A
WO20122166476	12/2011	WO	N/A
WO2014150514	12/2013	WO	N/A
WO2015006296	12/2014	WO	N/A
WO2015166487	12/2014	WO	N/A

OTHER PUBLICATIONS

Cutting Tool, Drill Bit, End Mill, Internet catalogue, <http://lzqtool.com/include/search.aspx?keycode=c-grade&type=1&language=en>, (Retrieved Feb. 7, 2018). cited by applicant

Tungsten Carbide Drills Mills & Burs, Internet catalogue, <http://chinatungsten.com/picture-bank/tungsten-carbide-drills.html>, (Retrieved Feb. 7, 2018). cited by applicant

MasterCut Tool Corp., Bur Series, Metric, (2018). cited by applicant

News & Notes, British Dental Journal, vol. 191, No. 7, pp. 410-411 (Oct. 13, 2001). cited by applicant

MasterCut Tool Corp., Bur Series, US, (2010). cited by applicant

Primary Examiner: Ramana; Anu

Background/Summary

PRIORITY CLAIM (1) In accordance with 37 C.F.R. 1.76, a claim of priority is included in an Application Data Sheet filed concurrently herewith. Accordingly, the present invention claims priority to U.S. Provisional Patent Application No. 63/142,716, entitled “SURGICAL IMAGE SYSTEM AND METHOD”, filed Jan. 28, 2021; also, the present invention is a Continuation-In-Part of U.S. Non-Provisional patent application Ser. No. 15/816,861, entitled “ROBOTIC SURGICAL SYSTEM”, filed Nov. 17, 2017, now U.S. Pat. No. 11,135,026, which claims priority

to U.S. Provisional Patent Application No. 62/423,677, entitled "ROBOTIC SURGICAL SYSTEM", filed Nov. 17, 2016. The contents of the above referenced applications are incorporated herein by reference.

FIELD OF THE INVENTION

(1) A high resolution surgical imaging system and method for real time use during surgery.

BACKGROUND OF THE INVENTION

(2) Surgical procedures, such as those performed on the spine, are well known in the art. The central nervous system is a vital part of the human physiology that coordinates human activity. It is primarily made up of the brain and the spinal cord. The spinal cord is made up of a bundle of nerve tissue which originates in the brain and branches out to various parts of the body, acting as a conduit to communicate neuronal signals from the brain to the rest of the body, including motor control and sensations. Protecting the spinal cord is the spinal, or vertebral, column. Anatomically, the spinal column is made up of several regions, including the cervical, thoracic, lumbar and sacral regions. Each of the vertebrae associated with the various spinal cord regions are made up of a vertebral body, a posterior arch, and transverse processes.

(3) While most people have fully functional spinal cords, it is not uncommon for individuals to suffer some type of spinal ailment or disorder which requires some type of surgical intervention. There are many different approaches taken to alleviate or minimize severe spinal disorders. One surgical procedure commonly used is a spinal fusion technique. Several surgical approaches have been developed over the years, and include the Posterior Lumbar Interbody Fusion (PLIF) procedure which utilizes a posterior approach to access the patient's vertebrae or disc space, the Transforaminal Lumbar Interbody Fusion (TLIF) procedure which utilizes a posterior and lateral approach to access the patient's vertebrae or disc space, and the Anterior Lumbar Interbody Fusion (ALIF) which utilizes an anterior approach to access the patient's vertebrae or disc space.

(4) In addition to the spinal implants or use of bone grafts, spinal fusion surgery often utilizes spinal instrumentation or surgical hardware, such as pedicle screws, plates, or spinal rods. Once the spinal spacers and/or bone grafts have been inserted, a surgeon places the pedicle screws into a portion of the spinal vertebrae and attaches either rods or plates to the screws as a means for stabilization while the bones fuse. Currently available systems for inserting the rods into pedicle screws can be difficult to use, particularly in light of the fact that surgeons installing these rods often work in narrow surgical fields.

(5) Moreover, since patients can vary with respect to their internal anatomy, resulting in varying curvatures of the spine, a surgeon may not always have a linear path, or may have anatomical structures that must be maneuvered around in order to properly insert the surgical rods into the pedicle screw assemblies. In addition to requiring surgical skill, difficulty in placing the rods correctly into the pedicle screws can result in unnecessary increases in the time it takes a surgeon to complete the surgical procedure. Prolonged surgery times increase the risk to the patient. More importantly, improperly aligning the rods and pedicle screw assemblies often results in post-surgery complications for the patient and requires corrective surgical procedures.

(6) Robotic surgery, computer-assisted surgery, and robotically-assisted surgery are terms for technological developments that use robotic systems to aid in surgical procedures. Robotically-assisted surgery was developed to help overcome the limitations of pre-existing minimally-invasive surgical procedures and to enhance the capabilities of surgeons performing open surgery.

(7) In the case of robotically-assisted minimally-invasive surgery, instead of directly moving the instruments, the surgeon uses one of two methods to control the instruments: either a direct telemanipulator or through computer control. A telemanipulator is a remote manipulator that allows the surgeon to perform the normal movements associated with the surgery while the robotic arms carry out those movements using end-effectors and manipulators to perform the actual surgery on the patient. In computer-controlled systems, the surgeon uses a computer to control the robotic

arms and its end-effectors, though these systems can also still use telemanipulators for their input. One advantage of using the computerized method is that the surgeon does not have to be present, but can be anywhere in the world, leading to the possibility for remote surgery. One drawback relates to the lack of tactile feedback to the surgeon. Another drawback relates to visualization of the surgical site. Because the surgeon may be remote or the surgery may be percutaneous, is it difficult for the surgeon to view the surgery as precisely as may be needed.

(8) In the case of enhanced open surgery, autonomous instruments (in familiar configurations) replace traditional steel tools, performing certain actions. The main object of such smart instruments is to reduce or eliminate the tissue trauma traditionally associated with open surgery.

(9) While robots are fully capable of repetitive tasks and work well in planned, routine settings, such environments are not always possible during a surgical procedure. In addition, robots are unintelligent in that they must be programmed to perform their functionality. However, this can be problematic when the environments they are programmed to function in are not static. As robotic systems become more prevalent in the surgical field, there exists a need for such robotic-assisted procedures to be performed safely and more intelligently, and capable of modifications in real time.

(10) To perform such surgery, it is typical to provide an image displayed on a screen to the surgeon showing the surgical site during surgery. Such images are sonograms and of low resolution, but are real time images. Before surgery, a high resolution image of the surgical site is generated, as with magnetic resonance imaging (MRI), computerized tomography (CAT scan), x-ray or the like, so the surgeon can determine how to perform the surgery and identify risk areas. These images provide vital information. However, these imaging methods cannot be performed during surgery to provide real time high resolution imaging. Instead, it is typical to provide the real time image with ultrasound to provide a live sonogram.

SUMMARY OF THE INVENTION

(11) The present invention provides apparatus, systems, and methods for use with robotically assisted surgery. The invention provides a system and method that will provide the surgeon with high resolution images in real time. The system and method utilize a plurality of fiducial points in the surgical site that are the same in a first high resolution image and the low resolution real time image generated during surgery, which allows enhanced viewing and tracking movement of a body structure at a surgical site. The tracked movement can be used to adjust movement of the robot and its surgical tool in real time.

(12) Accordingly, it is an objective of the invention to provide a system for use with robotically assisted surgery.

(13) It is a further objective of the invention to provide methods for use with robotically assisted surgery.

(14) It is a still further objective of the invention to provide methods of performing a robotically assisted surgical procedure using one or more robots.

(15) It is a still further objective of the invention to provide methods of performing a robotically assisted surgical procedure.

(16) It is even a further objective of the invention to program a computer to control movements of one or more robots used in the surgery.

(17) Other objectives and advantages of this invention will become apparent from the following description taken in conjunction with any accompanying drawings wherein are set forth, by way of illustration and example, certain embodiments of this invention. Any drawings contained herein constitute a part of this specification, include exemplary embodiments of the present invention, and illustrate various objects and features thereof.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a schematic illustration of portions of a robotic surgical system;
- (2) FIG. 2 is a schematic illustration of additional portions of a robotic surgical system;
- (3) FIG. 3 is a schematic illustration of a spinal surgical site in side and plan views; and
- (4) FIG. 4 is a schematic illustration of additional portions of a robotic surgical system.

DETAILED DESCRIPTION OF THE INVENTION

(5) Referring to FIGS. 1-3, a robotic surgical system **100** is illustrated. The robotic surgical system **100** generally includes a multi-axis robot **2**, a tool **4** (surgical tool which may be an oscillating tool assembly below) with an effector **5** on a distal end thereof, and an operator station **6**. The tool **4** can be of any suitable type, such as an oscillating tool and/or rotating tool. The multi-axis robot **2** includes a plurality of axes about which the oscillating tool **4** can be precisely maneuvered and oriented for surgical procedures. In a preferred, but non-limiting, embodiment, the multi-axis robot includes seven axes of movement. The axes of movement include the base axis **202**, generally centered within the base **200** and about which the first arm **204** rotates. The second axis **206** is substantially perpendicular to the first, base axis **202** and about which the second arm **208** rotates. The second arm **208** includes the third axis **210** about which the third arm **212** rotates. The third arm **212** includes the fourth axis of rotation **214**, which is oriented substantially perpendicular with respect to the first, base axis **202** and substantially parallel to the second axis **206**. The fourth arm **216** rotates about the fourth axis **214**. The fourth arm **216** includes the fifth axis **218** about which the fifth arm **220** rotates. The fifth arm **220** includes the sixth axis **222**, which includes the most available rotation about the sixth axis **222** for the wrist **224** of the robot. The wrist **224** carries the tool **4** and effector **5**, and has a seventh axis of rotation **228** for the cutting tool. The wrist **224** is at the distal end of the fifth arm **220**. It should be noted that each axis of rotation provides an additional freedom of movement for manipulation and orientation of the tool **4**. It should also be noted that while the multi-axis robot **2** is only illustrated with the tool **4**, the preferred embodiment is capable of changing the effector to a variety of tools that are selectively utilized to complete a particular surgery. Drives, not shown, are utilized to move the arms into their desired positions. The drives may be electric, hydraulic or pneumatic without departing from the scope of the invention. Rotational position can be signaled to a computer **230**, as with an encoder (not shown) associated with each arm **204**, **208**, **212**, **216**, **220** and other components having an axis of rotation. In the preferred embodiment, the drives are in electrical communication with the computer **230**, and may further be combined with a telemanipulator, or pendant (not shown). The computer **230** can be programmed to control movement and operation of the robot(s) **2** through a manually operated controller portion **231** or autonomously through software programming. Thus, the robotic movements can be programmed by inputting code which may include positions in a two dimensional or three dimensional area or by overlaying (drawing) tool paths **355** onto an image; whereby the computer determines the proper code to cause the robot to move according to the paths drawn on the image. These tool paths **355** may be two dimensional or three dimensional without departing from the scope of the invention as shown in FIGS. 3 and 4, including drilling milling, boring and the like. The computer **230** can have a primary storage device (commonly referred to as memory) and/or a secondary storage device that can be used to store digital information, such as images described herein. Primary and secondary storage are herein referred to as storage collectively, and can include one or both primary and secondary storage. The system **100** may further include sensors positioned along various places on the multi-axis robot **2**, which provide tactile feedback to the operator or surgeon **232**. The computer **230** is electrically connected or coupled to the multi-axis robot **2** in a manner that allows for operation of the multi-axis robot **2**, ranging from positions adjacent the robot to thousands of miles away. The computer **230** is preferably capable of accepting, retaining and executing programmed movements of the multi-axis robot **2** in a precise manner. In this manner, skilled surgeons can provide surgical care in areas,

such as battlefields, while the surgeon is safe from harm's way. The controller **231** can include a movement control input device **233**, such as a joy stick, keyboard, mouse or electronic screen **306**, which can be touch activated. The screen **306** can be part of the monitor **234**. Tool change commands can be input using the screen **306**.

(6) Referring to FIGS. **1**, **2**, and **4** the robotic surgical system **100** generally can include one or two multi-axis robot(s) **2**, with one carrying an ultrasound imaging system **300** and one carrying an effector, such as an oscillating tool **4**, both operably connected to the operator station **6** for control of operation. Typically, a surgeon **232** would utilize fluoroscopy or fluoroscopy in combination with computerized tomography (CT) scans or the like to create a high resolution image for use to evaluate surgery steps pre-surgery. Fluoroscopy images may be digitized, e.g. computerized, for viewing and storage when taken. The CT scans are performed prior to the surgery so the surgeon can identify landmarks within the patient **308**. The initially created high resolution image can be either 2D or simulated 3D. In order to provide real time images to the surgeon **232** during surgery, one of the robots **2** is fitted with an ultrasound imaging probe **302** of the imaging system **300**. The ultrasonic imaging probe **302** is electrically connected to an imaging system electronic controller **304** provided in the computer **230** which allows the operator to project the real-time images on the monitor **234**. The patient is fitted with a plurality of fiducial point devices **351**, for example, in vertebra **352** of a spine **353**, prior to initial imaging to produce the high resolution image; and that image is stored in the computer **230**. The initial high resolution image(s) and ultrasonic image(s) generated during surgery are stored in and recalled from the computer **230** storage and displayed on the monitor **234**. The fiducial point devices **351** remain fitted to the patient in their original locations during surgery. The computer will align the real time sonogram with the initial high resolution image using the fiducial points for co-registration of the fiducial point devices **351** to effect and maintain image alignment in real time, including the scale of the images. The computer will affect display of the high resolution combined image on the monitor **234** in place of the sonogram in its current orientation and position. In some embodiments, portions of the CT image are segmented from the whole image to allow the computer to move the segmented portions to match any movement detected in the real-time images taken. This also allows the computer to alter tool paths in real-time to match the movement of the bone structure of the patient. The segmentation also allows the combined image to be rotated for addition of tool paths, screw insertion or the like to aid in programming the movements, e.g. tool paths, of the surgical robot to complete the surgery. The monitor **234** may be positioned in the operator station **6** and/or within the operating room **310**. Thus, the operator can calibrate the robots positioning to correspond to the real-time ultrasonic image for completing the surgery while viewing the substituted high resolution image.

(7) Fiducial point devices **351** can also be used to assist in determining the position of a tool **4** relative to a patient **308**, and to assist in overlaying and aligning the high resolution and ultrasound images. Typically, for orthopedic surgery, fiducial point devices **351** are attached to a bone, as with a screw. Such fiducial point devices are available from Northern Digital, Inc. These devices can provide up to six degrees of monitored movement for which they provide feedback to the computer controlling the robot. In some cases, inserting more than one fiducial point device can be used to overcome interference and feedback issues that can be caused by items such as cellphones, electric motors and the like. By having redundant fiducial devices in the same bone or area of operation, one fiducial marker can be checked against the other to determine if one of the fiducial markers is sending erroneous information or failing to register movement. In this manner, electromagnetic or radio frequency sensors can be coupled with infra-red (IR) sensors or cameras to provide further cross checks to look for movement or improper positioning of the sensors.

(8) Referring generally to the Figures, a method of monitoring the location of bones for robotic surgery includes: inserting at least one electromagnetic sensor into a bone in a manner so that the electromagnetic sensor is visible when viewed in a computer tomographic scan; taking a

computerized tomographic scan of the bone and surrounding area so that the electromagnetic sensor(s) are visible in the computerized tomographic scan; storing the computerized tomographic scan on an electronic media that is computer readable; providing a surgical robot having an arm including at least two axes of movement, the robot electrically connected to a computer for controlling the movement of the robot axes, the computer having an input device for inputting commands for robot movement, the computer including a storage medium for storing and recalling electronic data; connecting a monitor to the computer to receive electrical signals therefrom for visualizing operations performed by the computer; connecting an ultrasound probe to the surgical robot; utilizing the computer input device to cause the surgical robot to move the ultrasound probe to a position to capture an image of the at least one electromagnetic sensor which may include surrounding tissues; storing the ultrasound image on the computer storage medium; causing the computer to show the computerized tomographic image on the monitor; recalling and overlaying the ultrasound image upon the computer tomographic image; and resizing the images to match each other using the electromagnetic sensor(s) visible in both images. The method may further include such features as including the step of segmenting the tomographic image and the ultrasound image so that only a portion of each image is visible on the monitor, wherein the tomographic image is segmented by a user to include bones chosen by the user, the ultrasound image segmented by the computer to match the user segmented tomographic image. The method may include the ability to rotate the segmented image on the monitor for viewing. These steps are operable from the computer which operates as an operating station for controlling operation of the robot and the image on the monitor. The images may be three dimensional images which allow the operator to overlay tool paths for the robot to follow during the surgery over the combined computer tomographic image and the ultrasound image. The image with the tool paths may then be utilized by the computer to construct electrical commands to cause the robot to move in accordance with the paths constructed by the operator. These codes may take several different forms known in the art for programming robots and computer numerical control (CNC) machining centers that operate in two dimensional and three dimensional spaces. An alternative method includes the steps of: inserting at least one electromagnetic sensor into a bone in a manner so that the electromagnetic sensor is visible when viewed in a radiographic scan; taking a computerized radiographic scan of the bone and surrounding area so that the electromagnetic sensor(s) are visible in the computerized radiographic scan; storing the computerized radiographic scan on an electronic media that is computer readable; providing a surgical robot having an arm including at least two axes of movement, the robot electrically connected to a computer for controlling the movement of the robot axes, the computer having an input device for inputting commands for robot movement, the computer including a storage medium for storing and recalling electronic data; connecting a monitor to the computer to receive electrical signals therefrom for visualizing operations performed by the computer; connecting an ultrasound probe to the surgical robot; utilizing the computer input device to cause the surgical robot to move the ultrasound probe to a position to capture an image of the at least one electromagnetic sensor which may include surrounding tissues; storing the ultrasound image on the computer storage medium; causing the computer to show the computerized radiographic image on the monitor; recalling and overlaying the ultrasound image upon the computerized radiographic image; and resizing the images to match each other using the electromagnetic sensor(s) visible in both images to assist in resizing the images. This method may further include the steps of segmenting the radiographic image and the ultrasound image so that only a portion of each image is visible on the monitor. Utilizing the segmentation of the tomographic image, a user may include bones chosen by the user, the ultrasound image segmented by the computer to match the user segmented radiographic image. The segmented image is rotatable on the monitor for viewing and overlayment of tool and cutter paths. The computer preferably operates as an operator station for controlling operation of the robot and the image on the monitor.

(9) All patents and publications mentioned in this specification are indicative of the levels of those skilled in the art to which the invention pertains. All patents and publications are herein incorporated by reference to the same extent as if each individual publication was specifically and individually indicated to be incorporated by reference.

(10) It is to be understood that while a certain form of the invention is illustrated, it is not to be limited to the specific form or arrangement herein described and shown. It will be apparent to those skilled in the art that various changes may be made without departing from the scope of the invention and the invention is not to be considered limited to what is shown and described in the specification and any drawings/figures included herein.

(11) One skilled in the art will readily appreciate that the present invention is well adapted to carry out the objectives and obtain the ends and advantages mentioned, as well as those inherent therein. The embodiments, methods, procedures and techniques described herein are presently representative of the preferred embodiments, are intended to be exemplary, and are not intended as limitations on the scope. Changes therein and other uses will occur to those skilled in the art which are encompassed within the spirit of the invention and are defined by the scope of the appended claims. Although the invention has been described in connection with specific preferred embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Indeed, various modifications of the described modes for carrying out the invention which are obvious to those skilled in the art are intended to be within the scope of the following claims.

Claims

1. A method for monitoring the location of bones for robotic surgery comprising: inserting at least one electromagnetic sensor into a bone; taking a computerized tomographic scan of the bone and surrounding area so that the at least one electromagnetic sensor is visible in the computerized tomographic scan; storing the computerized tomographic scan on an electronic media that is computer readable; providing a surgical robot having an arm including at least two axes of movement, the robot electrically connected to a computer for controlling the movement of the robot axes, the computer having an input device for inputting commands for robot movement, the computer including a storage medium for storing and recalling electronic data; connecting a monitor to the computer to receive electrical signals therefrom for visualizing operations performed by the computer; connecting an ultrasound probe to the surgical robot; utilizing the computer input device to cause the surgical robot to move the ultrasound probe to a position to capture an image of the at least one electromagnetic sensor which may include surrounding tissues; storing the ultrasound image on the computer storage medium; causing the computer to show the computerized tomographic image on the monitor; recalling and overlaying the ultrasound image upon the computer tomographic image; and resizing the images to match each other using the at least one electromagnetic sensor visible in both images.
2. The method for monitoring the location of bones for robotic surgery as claimed in claim 1 including the step of segmenting the tomographic image and the ultrasound image so that only a portion of each image is visible on the monitor.
3. The method for monitoring the location of bones for robotic surgery as claimed in claim 2 wherein the tomographic image is segmented by a user to include bones chosen by the user, the ultrasound image segmented to by the computer to match the user segmented tomographic image.
4. The method for monitoring the location of bones for robotic surgery as claimed in claim 1 wherein the segmented image is rotatable on the monitor for viewing.
5. The method for monitoring the location of bones for robotic surgery as claimed in claim 1 wherein the computer is an operating station for controlling operation of the robot and the image on the monitor.

6. The method for monitoring the location of bones for robotic surgery as claimed in claim 5 wherein the operator can overlay tool paths for the robot to follow during the surgery over the combined computer tomographic image and the ultrasound image.
 7. The method for monitoring the location of bones for robotic surgery as claimed in claim 6 wherein the computer constructs electrical commands to cause the robot to move in accordance with the paths constructed by the operator.
 8. The method for monitoring the location of bones for robotic surgery as claimed in claim 1 wherein the computer tomographic image is a three dimensional image.
 9. A method for monitoring the location of bones for robotic surgery comprising: inserting at least one electromagnetic sensor into a bone; taking a computerized radiographic scan of the bone and surrounding area so that the at least one electromagnetic sensor is visible in the computerized radiographic scan; storing the computerized radiographic scan on an electronic media that is computer readable; providing a surgical robot having an arm including at least two axes of movement, the robot electrically connected to a computer for controlling the movement of the robot axes, the computer having an input device for inputting commands for robot movement, the computer including a storage medium for storing and recalling electronic data; connecting a monitor to the computer to receive electrical signals therefrom for visualizing operations performed by the computer; connecting an ultrasound probe to the surgical robot; utilizing the computer input device to cause the surgical robot to move the ultrasound probe to a position to capture an image of the at least one electromagnetic sensor which may include surrounding tissues; storing the ultrasound image on the computer storage medium; causing the computer to show the computerized radiographic image on the monitor; recalling and overlaying the ultrasound image upon the computerized radiographic image; and resizing the images to match each other using the at least one electromagnetic sensor visible in both images to assist in resizing the images.
 10. The method for monitoring the location of bones for robotic surgery as claimed in claim 9 including the step of segmenting the radiographic image and the ultrasound image so that only a portion of each image is visible on the monitor.
 11. The method for monitoring the location of bones for robotic surgery as claimed in claim 10 wherein the tomographic image is segmented by a user to include bones chosen by the user, the ultrasound image segmented by the computer to match the user segmented radiographic image.
 12. The method for monitoring the location of bones for robotic surgery as claimed in claim 9 wherein the segmented image is rotatable on the monitor for viewing.
 13. The method for monitoring the location of bones for robotic surgery as claimed in claim 9 wherein the computer is an operating station for controlling operation of the robot and the image on the monitor.
 14. The method for monitoring the location of bones for robotic surgery as claimed in claim 13 wherein the operator can overlay tool paths for the robot to follow during the surgery over the combined computer tomographic image and the ultrasound image.
 15. The method for monitoring the location of bones for robotic surgery as claimed in claim 14 wherein the computer constructs electrical commands to cause the robot to move in accordance with the paths constructed by the operator.
-